Final Project : ECS 174

Image Colorization with CNNs

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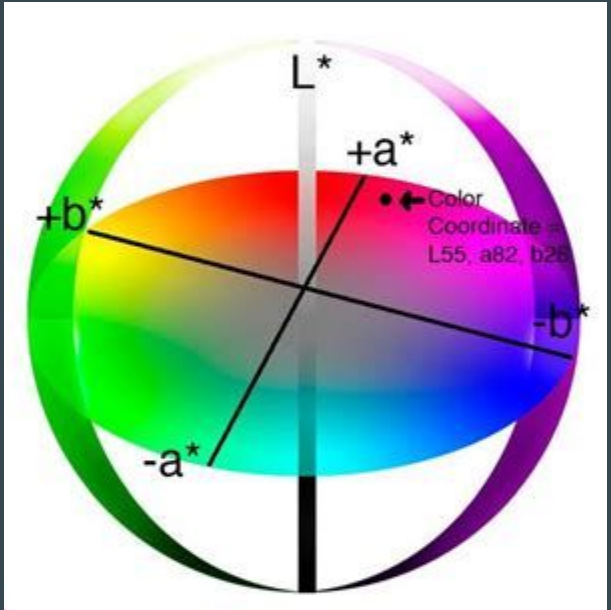
Introduction

- Goal: take a grayscale image and predict a plausible color version of the same scene
- Work in LAB color space instead of RGB
    - L = lightness, a/b = chrominance (color)
    - Separates brightness from color, easier to learn a,b from
- Model input: L channel only
- Model output: predicted a and b channels

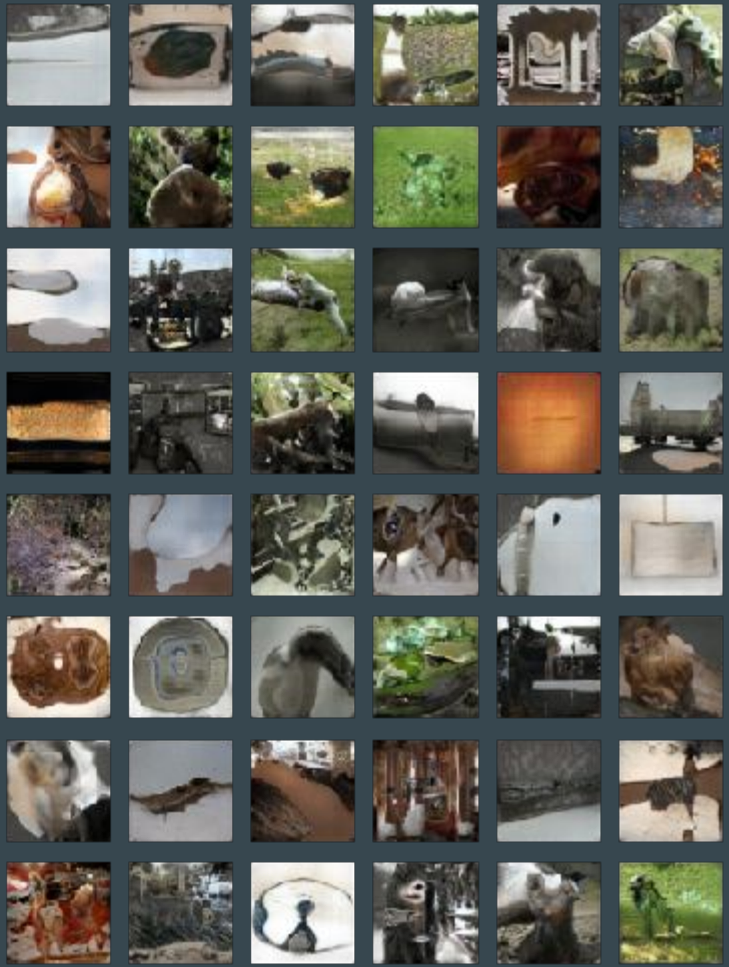
Motivation and Problem Setup

- Colorization sits between:
  - Low-level vision: edges, textures
  - High-level vision: object semantics (sky, grass, skin, etc.)
- Makes CNN concepts from class feel more “visual” than standard classification
- Input–output format:
  - Input: grayscale L channel (1 × H × W)
  - Output: predicted a,b channels (2 × H × W)
- Final image: combine L + (a,b) → convert LAB → RGB



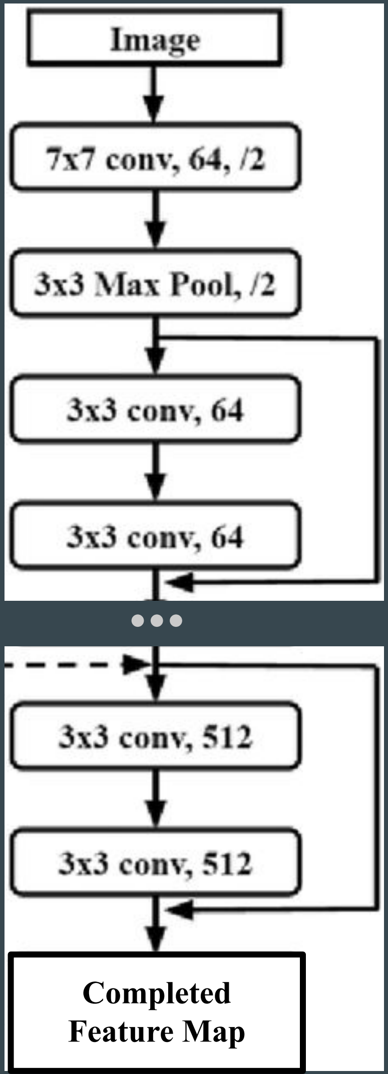
Tiny ImageNet-200 & Preprocessing

- Dataset: Tiny ImageNet-200
  - 200 classes, 500 train + 50 val images per class
  - Smaller, faster alternative to full ImageNet
- Preprocessing:
  - Resize all image to 224 x 224
  - Convert RGB → LAB using OpenCV
- Data augmentation
  - Random resized crop
  - Horizontal flip
  - Color jitter
- Lab normalization
  - Divide L, a, b by 255 → all channels in [0,1]
  - [0,1], sigmoid-style normalization was more stable than using tanh [-1,1].



Encoder - Decoder Architecture

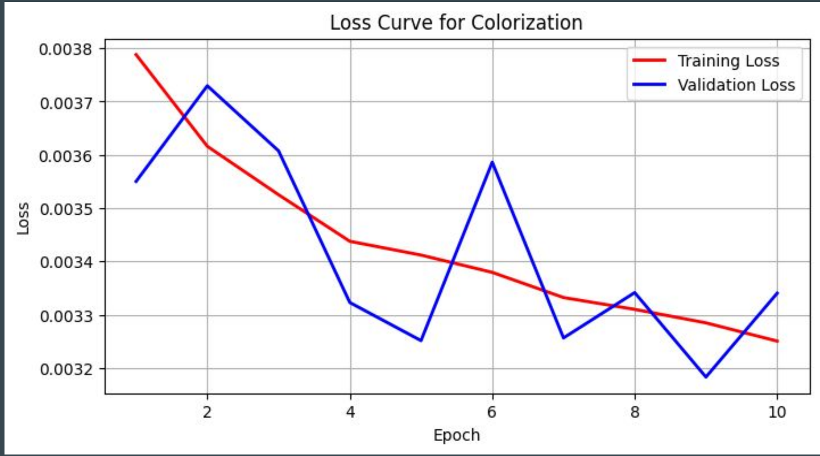
- Main model: convolutional encoder-decoder network
- Encoder:
    - Pretrained ResNet(IMAGENET1K_V1) as feature extractor
    - First conv layer modified to accept 1- channel L instead of RGB
    - Last 2 classification layer removed
- Decoder:
    - Transpose convolutions (learned non sampling)
    - Batch normalization + ReLU nonlinearities
    - Sigmoid at the end → outputs in [0,1]



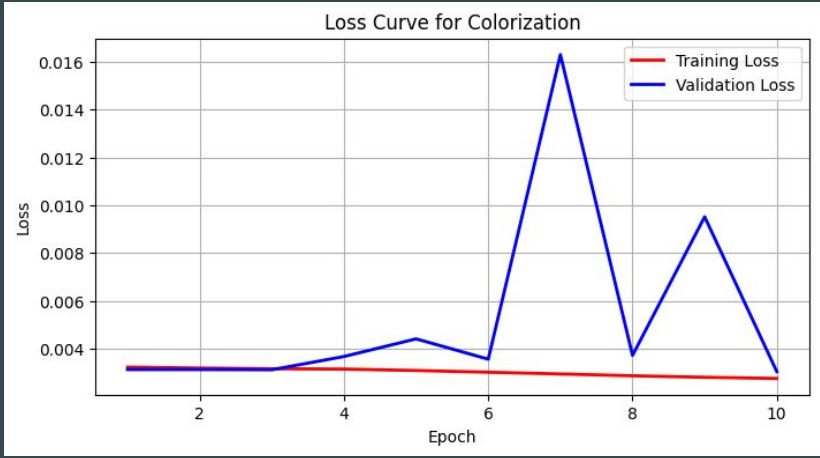
Training Setup and Evaluation

- Learned function:
  - fθ : L ∈ R^(1×H×W)
    → (â, b̂) ∈ R^(2×H×W)
- Training
  - Loss: MSE on a,b
  - Batch size: 16
  - Saved models every +10 epochs
    → up to 50 epochs
- Evaluation:
  - Quantitative: average test MSE on a, b
  - Qualitative: for sample images, show:
    - Original RGB, Grayscale L input, CNN colorization result

Epochs 1-10



Epochs 11-20



Results

- Model checkpoints: 10, 20, 30, 40 epochs
- Best performance:
  - 20-epoch model consistently best
  - Late models overtrained:
    - Washed-out colors
    - Oversaturation of red
- Test performance:
  - 20-epoch CNN: average test MSE ≈ 0.0030
- Visual results:
  - Outputs look similar to initial colored images
  - Subplots include:
    - Lab → RGB recon of original, Recolors from each saved model

Thank You